# BrainKAN Research Evidence Brief: Robust Statistical Evaluation of Learned Splines

**Author:** Xin Qi | **Repository:** https://github.com/xcklvydtj2024/BrainKAN-Explainability

**Environment:** Python 3.10.20 · PyTorch 2.14.0+cpu · PyG 2.8.0 · Deterministic seeds · Commit: `054ee7f`

## 1. Research Motivation & Core Methodological Problem

In functional neuroimaging graph neural networks, Kolmogorov–Arnold Networks (KANs) replace linear edge weights with learnable univariate B-splines $\Phi_{ij}(x) = \sum_p c_{ij,p} B_p(x)$, allowing direct visual inspection of edge-level transformations. In literature, researchers often interpret learned spline curvatures as evidence of in-vivo biological computation or task-specific cognitive mechanisms.

**The Foundational Methodological Problem:** *Do learned spline nonlinearities reflect task-specific information, or are they mathematical artifacts of operational metric definitions, evaluation domains, and non-convex optimization randomness?* We identify two primary unmodeled confounding factors:

1. **Reference-Domain Shifts:** Measuring curvature across shifted input supports creates apparent nonlinearity jumps purely from grid boundary extrapolation rather than model differences.
2. **Optimization Stochasticity:** Single initialization seeds can easily produce statistically significant conditional curvature differences that vanish or reverse across the broader optimization landscape. Conditional significance at Seed 42 ($p = 0.0060$) fails to generalize across 10 random seeds on the pooled domain ($\bar{\Delta}_{\text{pool}} = -0.00058$, positive in only 1/10 seeds).

## 2. Model Architecture: BrainKAN Edge-Spline Parameterization

The benchmark evaluates a specialized 2-layer Graph Kolmogorov–Arnold Network tailored for functional connectomes:

- **Layer 1 (Learnable Edge B-Splines):** For each directed functional edge $e = (u, v) \in \mathcal{E}$, message passing computes: $\Phi_e(x_u) = w_{\text{base},e} \cdot \text{SiLU}(x_u) + \sum_{k=1}^{G+k_o} c_{e,k} B_{k,k_o}(x_u)$, where $B_{k,k_o}$ denotes cubic B-spline basis functions ($k_o = 3$) parameterized over $G = 5$ grid intervals ($G + k_o = 8$ basis functions).
- **Exact Parameter Accounting:** For $|E| = 240$ edges, $C_{\text{in}} = 1$, and $C_{\text{out}} = 16$:
  - Spline coefficients: $240 \times 16 \times 1 \times 8 = \mathbf{30,720}$ parameters.
  - Base residual weights: $240 \times 16 \times 1 = \mathbf{3,840}$ parameters; Root self-connections: $16 \times 1 = \mathbf{16}$ parameters.
  - **Layer 1 Total: 34,576** parameters. Complete 2-layer BrainKAN model contains $P = \mathbf{589,779}$ trainable parameters.
- **Operational Metric (Deviation-From-Linearity, DFL):** For any learned function $f$ evaluated on discrete grid $D = \{t_1, \ldots, t_K\}$, DFL measures normalized variance unexplained by ordinary least-squares affine regression:

$$\text{DFL}(f; D) = \frac{\inf_{a,b} \sum_{k=1}^{K} |f(t_k) - (at_k + b)|^2}{\sum_{k=1}^{K} |f(t_k) - \bar{f}|^2} = \frac{\text{SS}_{\text{res}}}{\text{SS}_{\text{tot}}} = 1 - R^2$$

## 3. Dataset, Pairing Manifest, and Cross-Validation Integrity

- **Primary Cohort (HCP-WM):** 100 healthy young adults from the Human Connectome Project 3T Working Memory task-fMRI. Contrast: 2-Back (high cognitive load) vs. 0-Back (low cognitive load), comprising 200 total graphs (100 strictly paired subject observations).
- **Brain Graph Topology:** 48 cortical Regions of Interest (ROIs), 240 directed functional edges ($|E| = 240$).
- **Cross-Validation Integrity:** 5-fold CV with strict subject-level pairing (`data/subject_pairing_manifest.csv`). 0-Back and 2-Back scans for the same subject never cross fold boundaries. Fold reference grids $\mathcal{R}_k$ are computed exclusively from intra-fold training subjects (zero train-test leakage).
- **External Cohort (CHCP):** Chinese Human Connectome Project dataset utilized for cross-cohort transportability checks.

## 4. Core Experimental Design: Progressive Falsification Suite

The benchmark evaluates 5 pre-committed experiments crossing input shifts with design-consistent null models:

- **Exp 1 (Permutation Baseline):** Evaluates single-seed (`seed=42`) excess curvature on whole-graph average DFL over $N = 500$ sign-flip label permutations on $\mathcal{S}_{\text{pooled}}$ ($\Delta = +0.00162, p = 0.0060$). This global-graph test is distinct from exploratory edge-level FDR testing (where 0/240 edges reached significance).
- **Exp 2 (Synthetic Ground-Truth Calibration):** 3-condition simulation ($N = 200$) on known function $f^*(x) = x + 0.5\sin(2x)$ assessing DFL sensitivity to grid gauges.
- **Exp 3 (Reference Domain Control):** 5-fold CV on fMRI data contrasting pooled ranges against empirical common support $\mathcal{S}_{\text{common}} = \text{supp}(X_0) \cap \text{supp}(X_1)$.
- **Exp 4 (Capacity Sensitivity Analysis):** Multi-seed capacity scan across BrainKAN hidden dimensions ($d \in \{4, 8, 16\}$ across 5 random initializations).
- **Exp 5 (Multi-Seed Optimization Sensitivity Audit):** Multi-seed evaluation across $S = 10$ seeds $\times$ $B = 15$ permutations under **dual estimands**: (1) Label-Blind Pooled Support $\mathcal{S}_{\text{pooled}}$ ($\bar{\Delta}_{\text{pool}} = -0.00058$, 1/10 positive, showing initialization sensitivity); (2) Condition-Dependent Common Support $\mathcal{S}_{\text{common}}$ ($\bar{\Delta}_{\text{common}} = +0.00090$, 9/10 positive, exploratory observation limited by label-dependent grid selection).

## 5. Primary Empirical Findings Across Experimental Suite

| Experiment | Estimand & Target | Sample Budget | Effect Size / Statistic | p-value & Uncertainty |
|---|---|---|---|---|
| **Exp 1: Permutation** | Single-seed ($s = 42$) on $\mathcal{S}_{\text{pooled}}$ | $N = 500$ perms | $\Delta_{\text{pool}} = +0.00162$ | $p = 0.0060$ ($k = 2/500$). Significant conditionally at graph level. |
| **Exp 2: Synthetic** | Grid gauge sensitivity on $f^*(x)$ | $N = 200$ sims | Shifted $\Delta = +0.0816$ | SD = 0.0084 ($> 40\times$ inflation purely from grid shift). |
| **Exp 3: Common Support** | Cross-cohort transfer (HCP→CHCP) | 5-fold CV | $\Delta_{\text{common}} = -0.0003$ | 5-fold mean. Excess curvature collapses to zero. |
| **Exp 4: Capacity Scan** | BrainKAN hidden dims ($d = 4, 8, 16$) | 5 seeds × 3 dims | Baseline $T \approx 0.176 - 0.183$ | High test accuracy (90% – 93%). Baseline stable across widths. |
| **Exp 5: Multi-Seed (Pooled)** | Multi-seed on label-blind $\mathcal{S}_{\text{pooled}}$ | $S = 10$, $B = 15$ | $\bar{\Delta}_{\text{pool}} = \mathbf{-0.00058}$ | SD = 0.00100. 1/10 seeds positive. Single-seed finding is initialization-sensitive. |
| **Exp 5: Multi-Seed (Common)** | Multi-seed on common support $\mathcal{S}_{\text{common}}$ | $S = 10$, $B = 15$ | $\bar{\Delta}_{\text{common}} = \mathbf{+0.00090}$ | SD = 0.00055. 9/10 pos (exploratory seed $t = 5.19$). Grid selection limits confirmatory claim. |
| **Mechanistic Controls (Poly)** | Polynomial proxy on synthetic DGM | 100 null cohorts | FPR = 4.0% (sharp null) | Rejection rate = 7.0% in linear-shift stress test; 100% power at $\gamma = 1.0$. |
| **Mechanistic Controls (KAN)** | BrainKAN end-to-end audit | $\gamma \in \{0.0, 1.0\}$ | Planted DFL = 0.1910 | Unplanted DFL = 0.2579. Planted edges not recovered ($0.191 < 0.258$). |

## 6. Inferential Limitations & Methodological Boundary Conditions

To preserve rigorous scientific integrity, our findings are bounded by explicit inferential commitments:

1. **Single-Seed vs. Multi-Seed Generalizability:** Exp 1 demonstrates conditional statistical significance for Seed 42 ($p = 0.0060, B = 500$). However, on the identical pooled domain across $S = 10$ seeds, only 1/10 yields a positive difference ($\bar{\Delta}_{\text{pool}} = -0.00058$), confirming that the single-seed finding is sensitive to optimization initialization.
2. **Estimand Dependency of Common-Support Grids:** While 9/10 seeds yield positive excess curvature under common support ($\bar{\Delta}_{\text{common}} = +0.00090$), the grid is constructed conditionally on original task labels and evaluated against a fixed-grid null. This exploratory observation does not constitute a calibrated subject-level permutation test.
3. **Formally Discarding Mismatched Comparisons ($p = 0.00399$):** Naively comparing the 10-seed observed mean directly to a single-seed 500-permutation null yields an invalid pseudo-$p$-value ($p = 0.00399$) that is mathematically discordant with the formal test definition and is strictly discarded.
4. **Operational Residual Ratio vs. Differential Geometry:** DFL $= 1 - R^2$ is an operational metric of unexplained variance against affine regression over a discrete grid, **not** an intrinsic coordinate-invariant Riemannian curvature tensor.
5. **Linear-Shift Stress Test vs. Composite Null Calibration:** Under linear confounder shifts, exchangeability does not strictly hold. The observed 7.0% rejection rate demonstrates robustness against severe false-positive inflation in this stress test, but does not constitute formal composite null calibration.
6. **Non-Causal Attribution of Planted Edge Recovery:** BrainKAN exhibits lower DFL on planted edges (0.1910) than on unplanted edges (0.2579). These observations are consistent with possible effects of global pooling and excess model capacity, but do not isolate either mechanism causally without independent ablation.
7. **Empirical Fallback Numerical Guard Audit:** The common support guard (`if c_min >= c_max:`) was audited across all 5 folds × 48 ROIs (240 instances): 0/240 **triggered** (0.00%). All reported values reflect genuine non-empty overlaps.

## 7. Programmatic Provenance & Codebase Path Index

| Component | Repository Script / File Path | Role & Computational Output |
|---|---|---|
| **Architecture** | `models/kan.py` | Layer 1 B-spline EdgeConv (34,576 params) & BrainKAN (589,779 params). |
| **Baselines & Accounting** | `models/baseline.py`, `compute_baseline_params.py` | GCN/GAT baselines and analytical/runtime parameter counting assertions. |
| **Pairing Manifest** | `data/subject_pairing_manifest.csv` | 100 paired HCP subjects with verified 5-fold CV assignments. |
| **Domain & Disjointness** | `protocol/reference_domains.py`, `tests/test_reference_domains.py` | Label-blind/independent reference domains and strict fold partition assertions. |
| **Exp 1 (Permutation)** | `experiments/exp1_high_precision_permutation.py` | $N = 500$ single-seed permutation test → `results/exp1_high_precision/` |
| **Exp 2 (Synthetic)** | `experiments/exp2_synthetic_calibration.py` | Ground-truth gauge calibration ($N = 200$) → `results/exp2_synthetic/` |
| **Exp 3 (Common Support)** | `experiments/exp3_reference_domain_control.py` | Domain control & CHCP transfer → `results/exp3_ref_domain/` |
| **Exp 4 (Capacity Scan)** | `experiments/exp4_multi_seed_robustness.py` | Multi-seed hidden dim scan ($d \in \{4, 8, 16\}$) across 5 seeds. |
| **Exp 5 (Multi-Seed Audit)** | `experiments/hierarchical_inference_runner.py`, `exp5_runner.py` | 10-seed optimization sensitivity audit & dual-estimand evaluation. |
| **Mechanistic Controls** | `experiments/mechanistic_controls.py`, `tests/test_mechanistic_controls.py` | Identifiable polynomial benchmark, null calibration, and BrainKAN audit. |
| **Closed Audit Report** | `docs/audit_report.md` | Methodological audit closure with historical tracking and resolution commits. |